QUARTERLY EXAMINATION - 2026

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CLASS : 12 **PHYSICS** Reg.No : NKL

TIME : 3.00 Hours. MARKS : 70

PART – I

$15 \times 1 = 15$

Note: (i) Answer all the questions

(ii) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer

1. An electric field $\vec{E} = 10\,x\,\hat{\imath}$ exists in a certain region of space. Then the potential difference $V = V_o - V_A$, where V_o is the potential at the origin and V_A is the potential at x = 2 m is:

   a) 10 V b) - 20 V c) + 20 V d) - 10 V

2. If voltage applied on a capacitor is increased from V to 2V, choose the correct conclusion.

   a) Q remains the same, C is doubled b) Q is doubled, C doubled

   c) C remains same, Q doubled d) Both Q and C remain same

3. An electron enters between two horizontal plates separated by 2 mm and having a potential difference of 1000 V. The force on electron is

   a) 8×10^{-12} N b) 8×10^{-14} N c) 8×10^{9} N d) 8×10^{14} N

4. The temperature coefficient of resistance of a wire is 0.00125 per °C. At 20°C, its resistance is 1 Ω. The resistance of the wire will be 2 Ω at

   a) 800 °C b) 700 °C c) 850 °C d) 820 °C

5. Two resistors of 6 Ω and 9 Ω are connected in series to 120 V source. Power consumed by 6 Ω resistor is

   a) 1800 W b) 616 W c) 1500 W d) 384 W

6. A circular coil of radius 5 cm and 50 turns carries a current of 3 ampere. The magnetic dipole moment of the coil is

   a) 1.0 amp – m^2 b) 1.2 amp – m^2 c) 0.5 amp – m^2 d) 0.8 amp – m^2

7. A wire of length *l* carries a current *I* along the *Y* direction and magnetic field is given by $\vec{B} = \frac{\beta}{\sqrt{3}}(\hat{\imath} + \hat{\jmath} + \hat{k})T$. The magnitude of Lorentz force acting on the wire is

   a) $\sqrt{\frac{2}{3}}\beta Il$ b) $\sqrt{\frac{1}{3}}\beta Il$ c) $\sqrt{2}\beta Il$ d) $\sqrt{\frac{1}{2}}\beta Il$

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8. The magnetic susceptibility is negative for

a) Para magnetic material only b) Dia magnetic material only

c) Ferro magnetic material only d) Para and Ferro magnetic materials

9. When the current changes from +2A to −2A in 0.05 s, an emf of 8 V is induced in a coil. The co-efficient of self-induction of the coil is

a) 0.2 H b) 0.4 H c) 0.8 H d) 0.1 H

10. The instantaneous values of alternating current and voltage in a circuit are $i = \frac{1}{\sqrt{2}}\sin(100\pi t)\,A$ and $v = \frac{1}{\sqrt{2}}\sin(100\pi t + \frac{\pi}{3})\,V$. The average power in watts consumed in the circuit is

a) $\frac{1}{4}$ b) $\frac{\sqrt{3}}{4}$ c) $\frac{1}{2}$ d) $\frac{1}{8}$

11. A transformer is based on the principle of

a) Mutual induction b) Self induction c) Ampere's law d) Lenz's law

12. If the amplitude of the magnetic field is 3×10^{-6} T, then amplitude of the electric field for a electromagnetic waves is

a) 100 V m^{-1} b) 300 V m^{-1} c) 600 V m^{-1} d) 900 V m^{-1}

13. An e.m. wave is propagating in a medium with a velocity $\vec{v} = v\,\hat{i}$. The instantaneous oscillating electric field of this e.m. wave is along +y axis, then the direction of oscillating magnetic field of the e.m. wave will be along

a) - y direction b) - x direction c) + z direction d) - z direction

14. The power of a biconvex lens is 10 D and the radius of curvature of each surface is 10 cm. The refractive index of the material of the lens is

a) $\frac{4}{3}$ b) $\frac{9}{8}$ c) $\frac{5}{3}$ d) $\frac{3}{2}$

15. If the velocity and wavelength of light in air is V_a and λ_a and that in water is V_w and λ_w, then the refractive index of water is,

a) $\frac{V_w}{V_a}$ b) $\frac{V_a}{V_w}$ c) $\frac{\lambda_w}{\lambda_a}$ d) $\frac{V_a}{V_w}\frac{\lambda_a}{\lambda_w}$

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Answer any six questions. Question no. 17 is compulsory:

16. Define electric field. Mention its unit.
17. A copper wire of cross-sectional area 0.5 mm^2 carries a current of 0.2 A. If the free electron density of copper is $8.4 \times 10^{28}\ m^{-3}$ then compute the drift velocity of free electrons.
18. What is Seebeck effect?
19. A coil of a tangent galvanometer of diameter 0.24 m has 100 turns. If the horizontal component of Earth's magnetic field is 25×10^{-6} T then, calculate the current which gives a deflection of 60°.
20. State Ampere's circuital law.
21. What is meant by wattless current?
22. Give the uses of microwaves.
23. Why does sky appear blue?
24. List out the advantages of AC over DC.

PART – III $6 \times 3 = 18$

Answer any six questions. Question no. 30 is compulsory:

25. Obtain Gauss law from coulomb's law.
26. A sample of *HCl* gas is placed in a uniform electric field of magnitude $3 \times 10^4\ N\ C^{-1}$. The dipole moment of each *HCl* molecule is 3.4×10^{-30} Cm. Calculate the maximum torque experienced by each *HCl* molecule.
27. State Kirchhoff's current and voltage law.
28. What is mass spectrometer? Explain its principle.
29. How will induce an emf by changing the area enclosed by the coil?
30. The current flowing in the first coil changes from 2 A to 10 A in 0.4 s. Find the mutual inductance between two coils if an emf of 60 mV is induced in the second coil. Also determine the magnitude of induced emf in the second coil if the current in the first coil is changed from 4 A to 16 A in 0.03 s. Consider only the magnitude of induced emf.
31. Write down the properties of electromagnetic waves.
32. Derive the relation between f and R for spherical mirror.
33. What is critical angle? Obtain the equation for critical angle.

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PART – IV

$5 \times 5 = 25$

Answer all the questions:

34. a) Derive an expression for electrostatic potential due to an electric dipole.

(Or)

b) Describe the Fizeau's method to determine the speed of light.

35. a) Deduce the relation for the magnetic field at a point due to an infinitely long straight conductor carrying current using Biot Savart law.

(Or)

b) What is emission spectra? Explain the types of emission spectrum.

36. a) Explain the determination of unknown resistance using meter bridge.

(Or)

b) Derive an expression for phase angle between the applied voltage and current in a series RLC circuit.

37. a) Derive the relation for resultant capacitance, when capacitors are connected in series and in parallel.

(Or)

b) Discuss the conversion of galvanometer into an ammeter.

38. a) (i) State Joules law. Give its expression.

(ii) An electric heater of resistance 10Ω connected to 220 V power supply is immersed in the water of 1 kg. How long the electrical heater has to be switched on to increase its temperature from 30° C to 60°C. (Specific heat capacity of water is $s = 4200\ J\ kg^{-1}\ K^{-1}$)

(Or)

b) (i) What are the factors that depends on deviation of white light when passed through a prism?

(ii) The angle of minimum deviation for an equilateral prism 37°. Find the refractive index of the material of the prism.

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